

$$L' = L_i + L_e = \frac{1}{\pi} \left(\frac{1}{4} + \ln \frac{1}{a} \right) (\text{H/m})$$

List of Formula on Inductance :

- (i) Toroidal coil with N turns having rectangular cross-section with height h and inner and outer radii r_1 and r_2 .

$$L = \frac{\mu_0 \mu_r N^2 h}{2\pi} \ln \left(\frac{r_2}{r_1} \right) (\text{H})$$



- (ii) Toroidal coil with N turns having circular cross-section with area of cross-section S and mean radius ' r '.

$$L = \frac{\mu_0 \mu_r N^2}{2\pi r} S \text{ (H)}$$

- (iii) Solenoidal coil with N turns, S as the cross-sectional area and d as the length.

$$L = \frac{\mu_0 \mu_r N^2}{d} S \text{ (H/m)}$$

- (iv) Coaxial cable with inner radius a and outer radius b .

$$L = \frac{\mu_0}{8\pi} + \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right) \text{ (H/m)}$$

Here $\frac{\mu_0}{8\pi}$ is referred as the internal inductance per unit length.

and the second term is the external inductance per unit length.

- (v) Two parallel wires with separation d and radius of each wire a .

$$L = \frac{\mu_0}{4\pi} + \frac{\mu_0}{\pi} \ln\left(\frac{d}{a}\right) \text{ (H/m)}$$

Here again the first term is internal inductance while second term is external inductance.

Example 7.12.7 : Calculate the inductance of a 10m length of coaxial cable filled with a material for which $\epsilon_r = 18$, $\sigma = 0$, and $\mu_r = 80$ and having dimensions of $a = 1 \text{ mm}$, $b = 4 \text{ mm}$.

Solution : The inductance of a coaxial cable is

$$\begin{aligned} L &= \frac{\mu_0 \mu_r}{2\pi} \ln\left(\frac{b}{a}\right) \text{ (H/m)} = \frac{4\pi \times 10^{-7} \times 80}{2\pi} \ln\left(\frac{4}{1}\right) \\ &= 22.1807 \text{ } (\mu \text{ H/m}) \end{aligned}$$

For 10 m length,

$$L = 22.1807 \times 10 = 221.807 \text{ } (\mu \text{ H})$$

Example 7.12.8 : Calculate the inductance of a toroid formed by surfaces $r = 3 \text{ cm}$, $r = 5 \text{ cm}$, $Z = 0$ and $Z = 1.5 \text{ cm}$, wrapped with 5000 turns of wire, and filled with a magnetic material for which $\mu_r = 6$.

Solution : The inductance of a toroid is

$$\begin{aligned} L &= \frac{\mu_0 \mu_r N^2}{2\pi} \ln\left(\frac{r_2}{r_1}\right) \text{ (H)} = \frac{4\pi \times 10^{-7} \times 6 \times (5000)^2}{2\pi} \ln\left(\frac{5}{3}\right) \times (1.5 - 0) \\ &= 0.22987 \text{ (H)} \end{aligned}$$

Example 7.12.9 : Calculate the inductance of a solenoid of 2000 turns wound uniformly over a length of 500 mm on a cylindrical paper tube 40 mm diameter. The medium is air ($\mu = \mu_0$)

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Solution : The inductance of a solenoid with air medium -

$$L = \frac{\mu_0 \mu_r N^2}{d} S = \frac{4\pi \times 10^{-7} \times (2000)^2}{500 \times 10^{-3}} \frac{\pi}{4} \times (40 \times 10^{-3})^2$$

$$= 12.63 \text{ (mH)}$$

Example 7.12.10 : A toroidal coil of 500 turns is wound on a steel ring 0.5 m mean diameter and $2 \times 10^{-3} \text{ m}^2$ cross-sectional area. An excitation of 4000 A/m produces a flux density of 1T. Find the inductance of the coil.

Solution : We have

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$$L = \frac{\mu N^2}{2\pi r} \times S$$

The permeability is obtained by

$$\mu = \frac{B}{H} = \frac{1}{4000} = 0.25 \times 10^{-3} \text{ H/m}$$

Using the numerical data given,

$$\therefore L = \frac{0.25 \times 10^{-3} \times 500^2}{2\pi \times 0.25} \times 2 \times 10^{-3} = 80 \text{ mH}$$

Example 7.12.11 : Two coils are wound on a common magnetic circuit of 50 sq. cm in section and mean length of 100 cm. One coil has 80 turns and the other 700. Calculate the mutual inductance of the coils, if the relative permeability of the iron path is 1500.

If the current in the 80 turns coil changes from 5A to zero in 0.01 sec., find the induced emf in the second coil.

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Solution : In the previous example we learned that the mutual inductance

$$M = \frac{N_1 N_2}{l/\mu A} = N_1 N_2 \frac{\mu A}{l}$$

Using the numerical data given

$$M = 80 \times 700 \times \frac{4\pi \times 10^{-7} \times 1500 \times 50 \times 10^{-4}}{100 \times 10^{-2}} = 0.528 \text{ H.}$$

We have emf induced.

$$e = \frac{N_1 N_2}{l/\mu A} \frac{dI_1}{dt} = -M \frac{dI_1}{dt} = -0.528 \times \frac{(0 - 5)}{0.01} = 264 \text{ volts.}$$

Example 7.12.12 : Find the inductance per unit length of a co-axial cable neglecting the internal inductance. Radius of the inner conductor is 1 mm and radius of the outer hollow conductor is 3 mm. Assume $\mu_r = 1$.

MU - Dec. 2001

Solution : The inductance of a unit length of the co-axial transmission line is

$$L = \frac{\mu_0}{8\pi} + \frac{\mu_0}{2\pi} \ln \frac{b}{a} \text{ (H/m)} \quad \dots(\text{See Example 7.12.5})$$



The first term is the internal inductance per unit length of the inner conductor. The second term is the external inductance per unit length of the co-axial line. Neglecting internal inductance, we get

$$L = \frac{\mu_0}{2\pi} \ln \frac{b}{a} \text{ (H/m)}$$

Given : $b = 3 \text{ mm}; a = 1 \text{ mm}$ $\therefore L = \frac{4\pi \times 10^{-7}}{2\pi} \ln \left(\frac{3}{1} \right) = 0.22 \text{ } (\mu\text{H/m})$